

Hyperthyroidism and thyrotoxicosis 1

(NICE 2023, NG145, NICE 2023, NG12, Lancet 2024;403:768, JAMA 2023;330:1472, BMJ 2018;362:k2880, BMJ 2014;349:g5128, Lancet 2012;379:1155, NEJM 2019;381:749, BNF)



Red Whale

GEMS
Guidelines & Evidence Made Simple

Thyrotoxicosis in adults (this GEMS is NOT applicable to pregnant women/women trying to conceive)

Who to test (NICE)

If thyrotoxicosis suspected, check TSH, and, if abnormal, check T4 and T3 (most labs do this automatically).

If pituitary disease (secondary thyroid dysfunction), check TSH, and T4 +/- T3 if TSH low.

Repeat tests if symptoms worsen/new symptoms develop; wait 6w from initial test before repeat.

- Test if:
- Symptoms of hyperthyroidism.
 - New-onset atrial fibrillation.
 - Type 1 diabetes/other autoimmune disorder.

- Consider testing if:
- Depression.
 - Unexplained anxiety.

In children/young people, consider testing in all the above situations. Also consider testing if ANY of:

- Abnormal growth.
- Unexplained changes in behaviour/school performance.
- Do not offer testing solely because someone has type 2 diabetes.
- Do not test thyroid function at the time of an acute illness (can affect results).
- The use of biotin supplements (vitamin B7) can raise or lower TSH. If taking: stop and retest TSH several days later.

TSH reduced (usually <0.01mIU/L)

**TSH low, T3 and/or T4 raised:
THYROTOXICOSIS**

**TSH low, T3 and T4 normal:
SUBCLINICAL HYPERTHYROIDISM**

Consider referral if (NICE 2023, NG145): TSH <0.1mIU/litre on 2 tests at least 3m apart AND either:

- Symptoms of thyrotoxicosis OR
- Evidence of thyroid disease (e.g. goitre or positive thyroid antibodies).

Untreated subclinical hyperthyroidism (if decide not to refer/treat) (NICE 2023, NG145):

- Consider checking TSH every 6m. If TSH low, check T4 and T3.
- Consider stopping measuring TSH if it stabilises (2 similar measurements within the reference range 3–6m apart).

(Certain drugs can mimic the biochemical changes of subclinical hyperthyroidism: steroids, metformin and dopamine agonists.)

Risks: progresses to overt hyperthyroidism in 8% by 1y and 26% by 5y (others will return to normal). Progression is more likely if undetectable TSH or toxic multinodular goitre.

Benefits of treatment? Observational data shows association between subclinical hyperthyroidism and AF, heart failure, cardiac events, osteoporosis, dementia and death, BUT no evidence that treatment offers benefits.

1. Assess severity

Any symptoms/signs of thyrotoxic storm? (needs same-day referral)

- A thyroid storm is severe uncontrolled hyperthyroidism with multiorgan failure. Signs/symptoms include tachycardia, atrial fibrillation, fever, heart failure, vomiting, dehydration, jaundice, agitation, delirium and coma.
- Thyroid storm affects 0.2% of those with thyrotoxicosis, and can be precipitated by surgery, amiodarone use or stopping antithyroid drugs. Mortality is up to 17% if >60y.

2. Exclude thyrotoxic complications

Check BP, pulse and temperature, and look for signs of heart failure.

3. Look for red flags for malignancy

Rapidly-enlarging nodule, cervical lymphadenopathy, hoarseness (malignancy uncommon in functional thyroid nodules (i.e. those releasing excess thyroid hormone)).

4. Try to establish a cause

Look for extrathyroidal manifestations of Graves' disease (goitre, eye disease, pretibial myxoedema).

Examine for goitre: Is there a diffuse goitre? (suggestive of Graves' or multinodular goitre).

Is there an isolated lump?

Is thyroid gland tender? Fever and neck pain suggest thyroiditis: check CRP.

Think about possible drug causes, e.g. amiodarone. Ask about less obvious drugs: OTC iodine supplements (kelp), slimming pills, recent radiological contrast exposure.

5. Investigations

Check TSH receptor antibodies (TRAb) in all with thyrotoxicosis (confirms Graves' disease) (NICE).

Consider technetium scanning of the thyroid gland if TRAb negative (NICE) (likely a secondary care test).

If unexplained thyroid lump, consider suspected cancer pathway referral (NICE): see *Thyroid enlargement: GEMS*.

Hyperthyroidism and thyrotoxicosis 2

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Initial management in primary care (NICE 2023, NG145)

Transient thyrotoxicosis without hyperthyroidism: treat symptomatically, e.g. beta-blockers (unless contraindication). Anti-thyroid drugs are not effective (Lancet 2024;403:768).

- *NICE makes no further comment on the management of thyroiditis. We would suggest that identifying transient thyroiditis vs. hyperthyroidism may not always be clear, and it may be appropriate to liaise with secondary care.*

Hyperthyroidism:

- Refer to secondary care.
- Consider antithyroid drugs plus symptomatic management, e.g. beta-blockers (unless contraindication) while awaiting specialist assessment/further treatment.

Antithyroid drugs: things to consider if starting while awaiting specialist review

- **Carbimazole usually treatment of choice.** Propylthiouracil should be used instead of carbimazole if adverse reaction to carbimazole, pregnant/planning a pregnancy in the next 6 months, history of pancreatitis (NICE 2023, NG145).
- **Baseline tests required before starting:** FBC and LFTs.
- **Risks of antithyroid drugs:**
 - **Major risks: agranulocytosis and liver dysfunction (see below).**
 - **All patients need written information advising them of the risks of agranulocytosis and to get an urgent FBC if they develop a sore throat or fever.**
 - **Minor risks: rash, arthralgia: occur in 1–5% of patients** (commoner at higher doses).
- **Carbimazole dose for adults:** 15–40mg daily; continue until the patient becomes euthyroid (usually after 4 to 8 weeks), then reduce to 5–15mg daily. After that, reduce gradually and stop. Therapy usually given for 12 to 18 months. Higher doses should be prescribed under specialist supervision only (BNF).
- **Remission (= normal thyroid function 12m after stopping medication):** achieved in 50% in trials/30% in real life.

Dose adjustment of non-thyroid medication: certain drugs may need dose adjustment in hyperthyroidism:

- Warfarin: may require a lower dose in hyperthyroidism.
- Propranolol, steroids and digoxin: metabolism increased: may require a higher dose in hyperthyroidism.

Secondary care management of hyperthyroidism

In secondary care, long-term treatment may be with antithyroid drugs, radioactive iodine or surgery.

Who receives which option is a secondary care decision individualised to each patient.

For each of these options, monitoring of bloods is required; this may be carried out by primary or secondary care.

Antithyroid drugs	Monitoring (NICE 2023, NG145): Baseline tests: FBC and LFTs prior to starting. Monitoring in first 6–12m: • Check TSH, T4/T3 every 6w until normal. • Once TSH normal: check TSH every 3m. (FBC/LFT monitoring not required unless concern about agranulocytosis/hepatotoxicity) After antithyroid drugs stopped: check TSH (+/-T4 and T3) within 8w, then every 3m for 1y. Long-term monitoring: TSH annually.	Major side-effects: • Agranulocytosis (5 per 1000 patients): all patients need written information advising them to get an urgent FBC if they develop a sore throat or fever. If agranulocytosis develops, stop antithyroid drugs (do not restart) and refer to specialist. • Hepatotoxicity: ◦ Carbimazole: 1 per 250 patients. ◦ Propylthiouracil: 1 per 37 patients.
Radioactive iodine	Monitoring in first 6–12m (NICE 2023, NG145): • Check TSH, T4 and T3 every 6w for the first 6m, until TSH within normal range. • If TSH in reference range 6m after treatment, check at 9 and 12m. Long-term monitoring: TSH every 6m.	• Results in euthyroidism or hypothyroidism in 70–90%. • Contraindicated in pregnancy and breastfeeding (and pregnancy must be avoided for 6m after treatment). • Must follow radiation protection guidelines (e.g. avoid close contact with children or pregnant women for 3w). • Can worsen Graves' eye disease if present.
Surgery	Total thyroidectomy (NICE 2023, NG145): • Need lifelong levothyroxine post-surgery. If hemithyroidectomy (NICE 2023, NG145): • Check TSH and T4 at 2 and 6m post-surgery. • Long-term monitoring: TSH annually.	• Invasive and expensive. • Leaves a scar. • Results in permanent hypothyroidism. • Surgical complications, e.g. recurrent laryngeal nerve palsy, hypoparathyroidism (occurs in 4–6%).

We make every effort to ensure the information in these pages is accurate and correct at the date of publication, but it is of necessity of a brief and general nature. The information presented herein should not replace your own good clinical judgement, or be regarded as a substitute for taking professional advice in appropriate circumstances. In particular, we suggest you carefully consider the specific facts, circumstances and medical history of any patient, and recommendations of the relevant regulatory authorities. We also suggest that you check drug doses, potential side-effects and interactions with the British National Formulary. Save insofar as any such liability cannot be excluded at law, we do not accept any liability for loss of any type caused by reliance on the information in these pages. September 2024. For full references see the relevant Red Whale articles.